

PROTOCOL DEVIATION ADDENDUM

Offspring age classification for descriptive and sensitivity analyses

Protocol: PCNE protocol v4 (11 November 2025)
Decision date: 18 August 2026
Status: Approved for the v10 codebook and analysis preparation
Scope: Post-extraction classification; no change to eligibility, extraction, or the primary continuous age moderator

1. Deviation and rationale

The registered protocol specifies offspring age in days and identifies age or age class as a possible biological moderator, but it does not prespecify species- and sex-specific thresholds for an age-class variable. This addendum records the threshold rule approved after data extraction and before effect-size modelling.

Primary analysis decision: offspring_age_d remains the continuous age moderator. The added offspring_age_class variable is restricted to descriptive summaries and sensitivity analyses. It will not replace offspring_age_d in the primary moderator model.

This is a classification decision rather than a correction to author-reported data. The reported or extracted offspring age remains unchanged.

2. Operational age classes

Thresholds follow the working postnatal stages summarized by Bell (2018). For the planned three-level variable, Bell’s peripubertal and late-adolescent stages are combined as adolescent. Adult begins after the upper bound of Bell’s late-adolescent stage.

Species	Sex	Juvenile	Adolescent	Adult
Rat	Female	PND 18-34	PND 35-70	PND >70
Rat	Male	PND 18-41	PND 42-70	PND >70
Mouse	Female	PND 18-27	PND 28-60	PND >60
Mouse	Male	PND 18-29	PND 30-60	PND >60

Source: Bell 2018, Table 1; DOI 10.1210/en.2018-00220.

Mixed-sex rule: classify a mixed-sex row only when the male and female rules give the same class at the recorded age. Otherwise code transitional_unclear and review it in sensitivity analysis.

Boundary rule: the present database contains no offspring ages below PND 18. If a future row falls below PND 18, do not force it into juvenile; record it for a new prespecified decision. Bell notes that exact timing can vary by strain and environmental context.

Application to PCN v10: 52 effect sizes are juvenile, 78 adolescent, and 77 adult. No current row is transitional_unclear.

3. Codebook implementation

The clean extraction file adds offspring_age_class immediately after offspring_age_d. It also renames gest_stage_ex_new to gest_window and uses the approved labels early_mid, mid_late, late, and full_gestation.

measurement_variable is retained unchanged because it identifies the specific behavioural outcome. No independent-arm indicator is added because the control and exposed animals are distinct for every comparison; repeated measurements and shared groups will instead be represented by clustering identifiers in the derived analysis data.

The original statistical names c_se and ex_se are restored for author-reported or directly extracted standard errors. Analysis-ready statistics use the _analysis suffix and contain the final values selected for modelling, including reproducible derivations where the author did not report the required statistic.

4. Standard-error provenance fields

c_se: Control-group SE reported by the authors or directly extracted from the source; preserved unchanged.

c_se_analysis: Final control-group SE selected for analysis. It equals c_se when reported and accepted; otherwise it is derived reproducibly.

c_se_derivation: Method used to obtain or validate c_se_analysis.

c_se_source: Text provenance for c_se_analysis: the exact paper table/figure, author file/email, or input source and derivation used. It is not a numeric SE and does not replace c_se.

The experimental-arm variables ex_se, ex_se_analysis, ex_se_derivation, and ex_se_source use the same definitions.

5. References retained for citation

Bell, M. R. (2018). Comparing postnatal development of gonadal hormones and associated social behaviors in rats, mice, and humans. *Endocrinology*, 159(7), 2596-2613. <https://doi.org/10.1210/en.2018-00220>

Moore, E. M., Linsenbardt, D. N., Melón, L. C., & Boehm, S. L., II. (2011). Ontogenetic differences in adolescent and adult C57BL/6J and DBA/2J mice: Anxiety-like, locomotor, and consummatory behaviors. *Developmental Psychobiology*, 53(2), 141-156. <https://doi.org/10.1002/dev.20501>

Bell (2018) is the threshold authority for the operational rule. Moore et al. (2011) is retained as supporting context for mouse adolescence and behavioural testing across development.